

WAREN, Germany

WMO: 102680

Lat: 53.5197N

Lon: 12.6653E

Elev: 214

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	16.9	6.2	7.6	14.1	11.7	10.0	21.1	26.3	41.1	22.7	42.3	5.2	80	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	84.6	65.9	81.6	65.0	78.2	64.0	69.0	79.1	67.3	76.6	65.9	74.2	6.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	96.4	71.4	64.4	91.3	70.3	62.9	86.4	68.7	33.2	79.3	31.9	76.8	30.7	74.1	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 20.9	17.9	15.7	DB	7.7	90.9	7.5	3.4	2.3	93.4	-2.1	95.4	-6.3	97.3	-11.7	99.8	(4)
(5)			WB	7.1	73.7	7.4	3.8	1.8	76.4	-2.5	78.6	-6.6	80.7	-12.0	83.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	32.4	34.1	38.7	48.0	56.1	61.2	65.8	65.7	58.3	49.3	40.9	33.9		
	DBStd	13.79	8.37	7.84	6.85	7.34	6.27	5.65	5.59	5.27	5.41	6.33	6.01	7.87		
	HDD50	2352	547	446	352	122	15	0	0	0	3	91	275	501		
	HDD65	6129	1012	866	814	511	283	143	57	57	210	488	723	965		
	CDD50	1909	0	0	3	62	206	336	491	487	253	68	3	0		
	CDD65	209	0	0	0	2	8	28	83	78	10	0	0	0		
	CDH74	1773	0	0	0	17	105	269	737	569	75	1	0	0		
	CDH80	443	0	0	0	1	15	69	215	137	6	0	0	0		
(14) Wind	WSAvg	7.4	8.1	8.4	8.0	7.3	6.8	6.8	6.4	6.5	6.6	7.8	7.4	8.5		
(15) Precipitation	PrecAvg	23.0	1.9	1.5	1.6	1.3	2.0	2.4	2.6	2.3	2.0	1.7	1.8	2.0		
	PrecMax	28.8	3.6	3.5	3.7	2.7	3.7	4.7	11.3	5.9	5.9	3.4	3.9	4.2		
	PrecMin	15.9	0.0	0.3	0.4	0.1	0.4	0.4	0.4	0.8	0.5	0.8	0.2	0.4		
	PrecStd	3.4	1.0	0.7	0.8	0.7	0.9	1.0	2.0	1.1	1.2	0.7	0.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	53.0	61.7	76.5	81.9	87.0	89.9	87.5	80.4	68.0	55.8	51.8		
		MCWB	49.2	49.9	50.7	59.2	63.0	67.0	68.0	67.6	64.0	61.1	53.0	50.2		
	2%	DB	48.0	49.8	56.5	69.0	76.8	80.8	85.4	84.0	76.1	63.8	53.0	48.6		
		MCWB	46.2	47.0	48.5	55.5	61.8	65.2	67.1	65.0	62.8	58.3	50.5	46.8		
	5%	DB	45.4	47.3	52.1	64.9	72.2	76.3	81.9	80.8	71.8	60.6	51.1	45.8		
		MCWB	43.7	44.8	46.6	52.9	59.6	62.9	65.6	64.8	61.1	56.5	49.0	44.4		
	10%	DB	42.9	44.6	48.6	60.8	68.6	72.5	77.9	76.7	67.8	58.0	48.9	43.7		
		MCWB	41.3	42.3	44.5	51.4	57.7	61.0	64.3	64.3	59.5	55.2	47.1	42.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	50.7	52.7	60.8	66.7	70.4	71.3	71.2	66.6	62.7	53.6	50.9		
		MCDB	51.3	52.5	58.0	71.5	77.0	80.9	82.5	81.1	75.5	67.8	54.7	51.3		
	2%	WB	46.5	47.5	49.9	57.3	63.7	67.1	69.2	68.6	64.5	59.3	51.1	47.1		
		MCDB	47.9	49.0	54.4	66.5	73.4	76.6	80.2	77.0	72.0	61.8	52.5	48.4		
	5%	WB	43.8	45.2	47.0	54.7	61.2	64.8	67.5	66.9	62.7	57.5	49.1	44.6		
		MCDB	45.1	46.9	51.2	62.4	69.3	72.6	78.0	74.7	69.4	59.9	50.6	45.6		
	10%	WB	41.5	42.4	45.0	52.2	59.0	62.6	65.9	65.5	60.9	55.4	47.3	42.5		
		MCDB	42.7	44.3	48.1	59.1	66.4	70.3	74.8	73.0	66.6	57.8	48.6	43.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	9.2	12.5	16.5	17.3	16.7	16.4	14.7	11.5	7.5	6.6			
		MCDBR	8.5	10.7	18.6	23.3	24.3	24.0	23.7	22.8	21.3	14.2	8.6	7.5		
	5% WB	MCWBR	8.0	8.3	11.7	11.9	11.7	10.9	8.7	8.5	10.0	9.2	6.8	6.9		
		MCWBR	8.5	10.4	16.2	20.6	20.9	21.2	20.6	19.1	18.3	13.3	8.1	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.338	0.372	0.402	0.398	0.396	0.415	0.411	0.377	0.360	0.336	0.307		
		taud	2.318	2.313	2.252	2.237	2.287	2.313	2.279	2.303	2.379	2.392	2.372	2.315		
	Ebn at Noon		206	239	254	262	270	271	263	256	251	228	196	184		
		Edh at Noon	19	26	34	39	39	38	39	36	30	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		190	400	775	1293	1616	1701	1585	1337	978	526	228	130		
	RadStd		22	56	89	183	194	134	221	109	105	70	25	11		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page